

Ideassion Technology Solutions

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Domain: Backend(Springboot)

Section A – Theory

1.Explain the difference between DBMS and RDBM.

DBMS	RDBMS
DBMS stands for database management system.	RDBMS stands for relational database management system
Stores data as a file	Stores data in related tables using relationships.
Does not strictly support relationships	Supports relationships using primary and foreign key
Less secure	More secure and support data integrity
File System, XML Database	PostgreSQL, MySQL, Oracle

2.What is a Primary Key? Why is it important?

A Primary Key is a Column that uniquely identifies each row in table.It is used to prevent duplicate and Null values, maintain data integrity and establish relationships between table using foreign keys.

3.What is a Foreign Key?

A Foreign key is a column in one table that references the primary key of another table.It is used to create a relationship between two tables and maintain referential integrity,ensuring that the data in related tables remain consistent.

4. Explain the difference between a Row and a Column.

Row	Column
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Represents a single record.	Represents an attribute or field.
Horizontal	Vertical
Contains values for one entity.	Contains similar data.

5. Name any four SQL commands and explain their purpose.

SQL Command	Purpose
SELECT	Retrieves data from the table
INSERT	Add new records into the table
DELETE	Remove records from table
UPDATE	Modifies Existing records

Section B – SQL Queries

Employee Table:

Empid	Name	Department	Salary
101	John	IT	50000
102	Priya	HR	45000
103	Raghul	IT	60000
104	Sneha	Finance	55000
104	Arun	IT	70000

1. Display all employees.

```
SELECT * FROM Employee;
```

2. Display employees from the IT department.

```
SELECT * FROM Employee
```

```
WHERE Department = 'IT';
```

3. Display employees whose salary is greater than 55,000.

```
SELECT * FROM Employee
```

```
WHERE Salary > 55000;
```

4. Update John's salary to 52,000.

```
UPDATE Employee
```

SET Salary = 52000

WHERE Name = 'John';

5. Delete the employee with EmpId = 102.

DELETE FROM Employee

WHERE EmpId = 102;

6. Display all unique department names.

SELECT DISTINCT Department

FROM Employee;

Section C – Practical

1. Create the Student Table

CREATE TABLE Student (

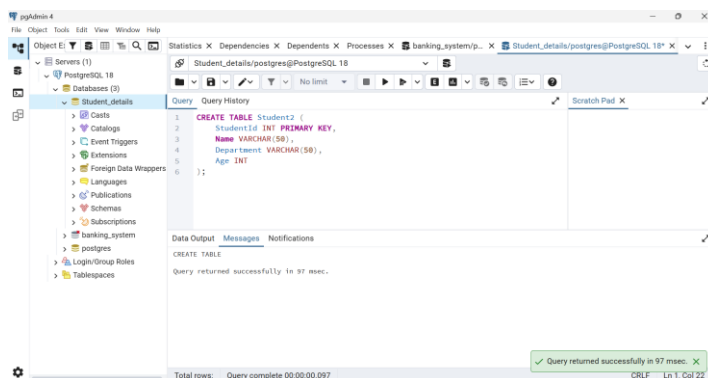
StudentId INT PRIMARY KEY,

Name VARCHAR(50),

Department VARCHAR(50),

Age INT

);



2. Insert at least 3 records

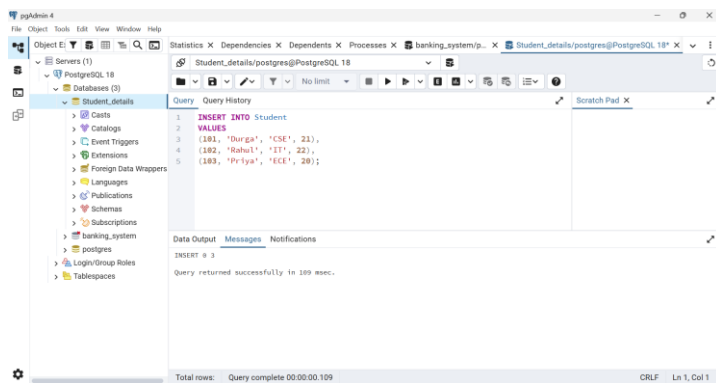
INSERT INTO Student

VALUES

(101, 'Durga', 'CSE', 21),

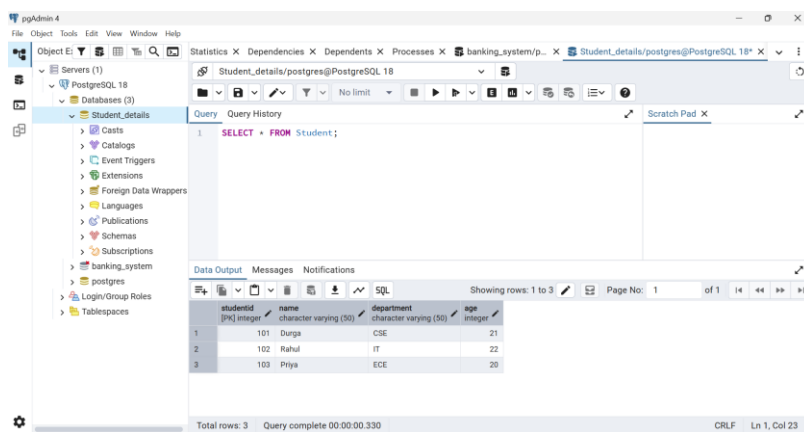
(102, 'Rahul', 'IT', 22),

(103, 'Priya', 'ECE', 20);



3. Display all records

SELECT * FROM Student;

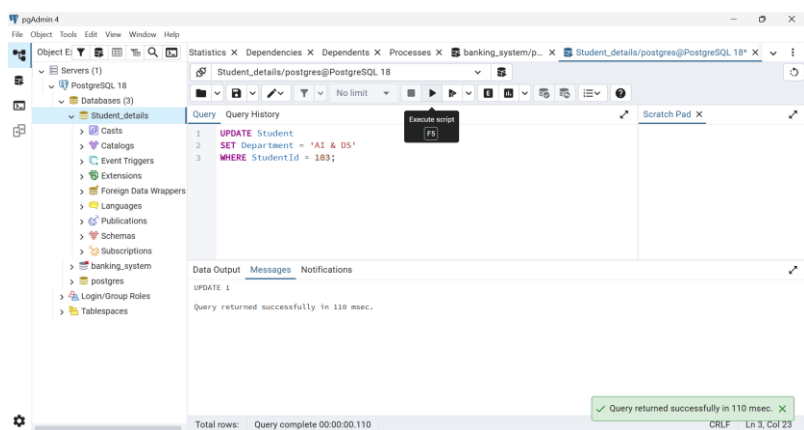


4. Update one student's department

UPDATE Student

SET Department = 'AI & DS'

WHERE StudentId = 103;



5. Delete one record

DELETE FROM Student

WHERE StudentId = 102;

